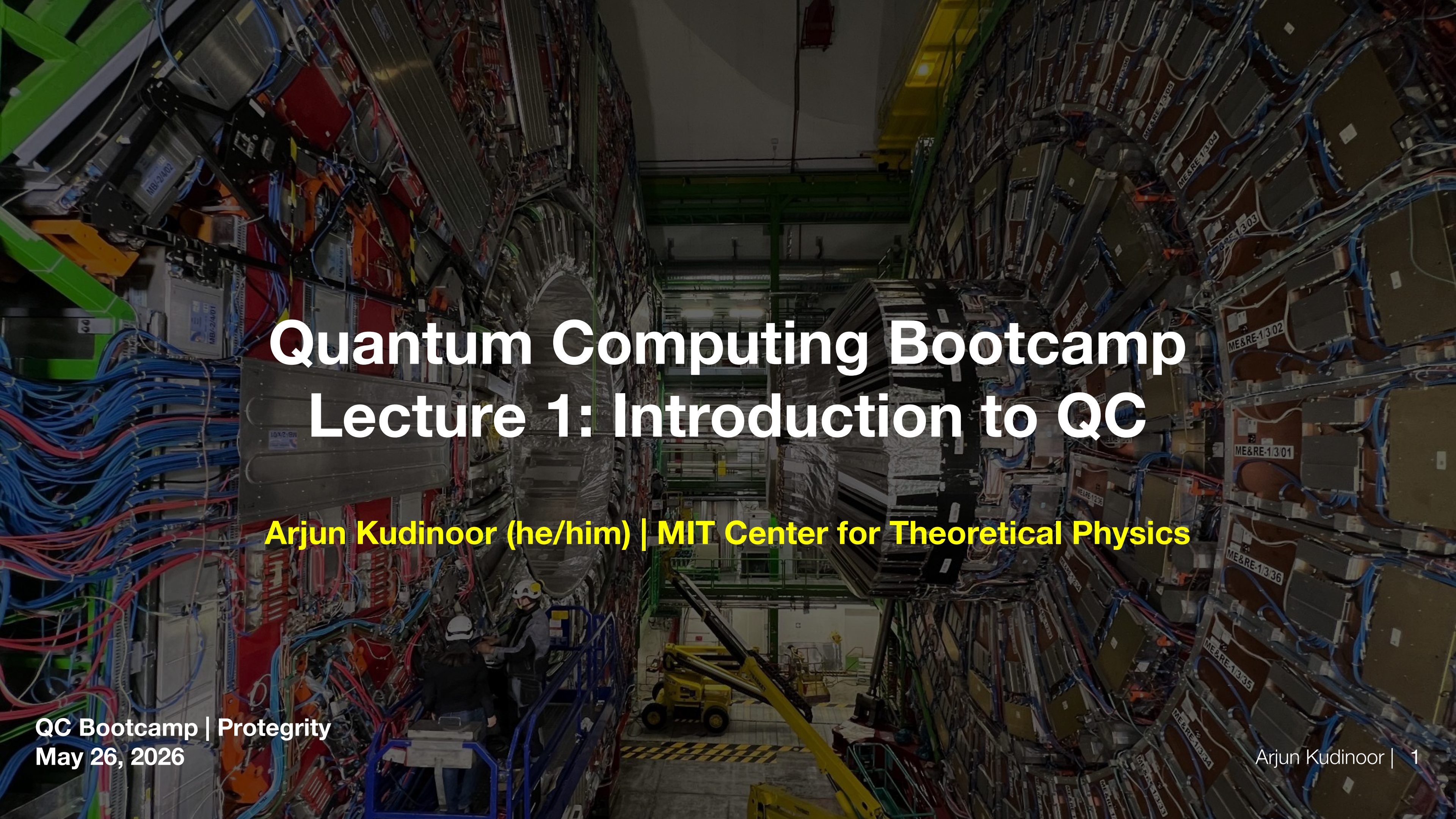


Quantum Computing Bootcamp
Lecture 1: Introduction to QC

Arjun Kudinoor (he/him) | MIT Center for Theoretical Physics

QC Bootcamp | Protegrity
May 26, 2026

A Little Bit About Me

PhD Student at the MIT Center for Theoretical Physics - a Leinweber Institute

BA at Columbia University (CC ‘23), where I co-taught and co-authored curriculum for the Quantum Computing course taken by physics majors

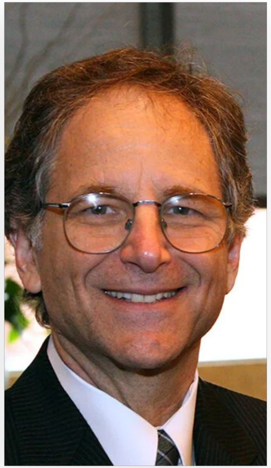
Advisor at Protegrity since March 2025: I help build knowledge of, expertise in, and implementation of quantum computing and quantum-safe encryption

CRYPTOGRAPHIC AND QUANTUM REVOLUTIONS

1977: R,S, and A establish RSA

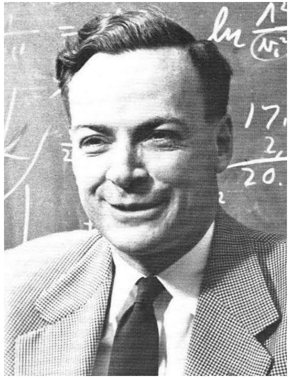
Ron Rivest

Adi Shamir



Leonard Adleman

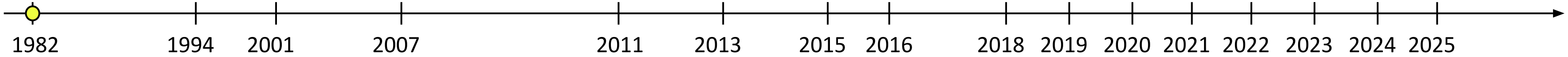
1982: Feynman’s Idea for Quantum Computation



Richard Feynman
(Caltech)

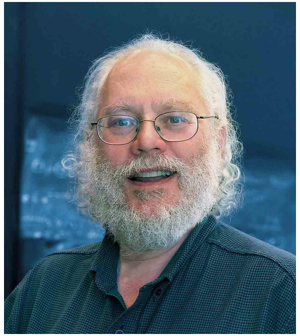
Richard Feynman proposed a new type of computer, a quantum computer

- **“It is impossible to represent the results of quantum mechanics with a classical universal device”**
- **The number of degrees of freedom in a system of N quantum mechanical particles scales exponentially with N**
- **To truly simulate quantum systems, you need a quantum computer**



[1] Feynman’s Keynote Address

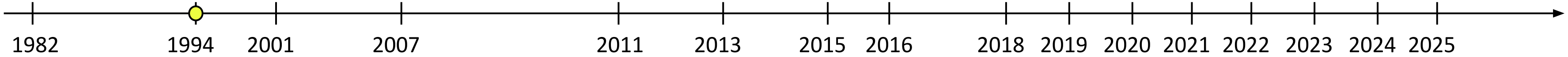
1994: Shor's Factoring Algorithm



Peter Shor (MIT)

Peter Shor formulated the Shor Algorithm. It factorizes integers exponentially faster than the fastest known classical algorithm

- **Threatens factorization-based cryptography**
- **Threatens elliptic curve discrete log cryptography**
- **Threatens the encryption of almost all information sent over the internet**



[2] Peter Shor's Algorithm

2001: Shor’s Algorithm is realized on a Quantum “Device”

Ike Chuang (MIT)

Lieven Vandersypen (QuTech @ Delft)

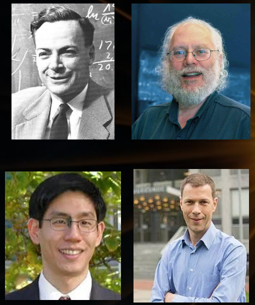
Isaac Chuang, Lieven Vandersypen, and IBM factorize 15

- **Engineered a special molecule ⇒ a tiny quantum computer**
- **Used nuclear magnetic resonance to modify spin states and perform computations**
- **First physical demonstration of Shor’s algorithm!**

The first-step to breaking factorization-based encryption, like RSA

[3] IBM’s Blog

The Big Bang of Quantum Computing



QUANTINUUM

IBM Quantum

Microsoft Azure

rigetti

IONQ

D-Wave The Quantum Computing Company™

QCi QUANTUM COMPUTING INC

aws Amazon Braket

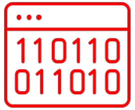
Google Quantum AI

QuEra Computing Inc.

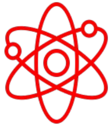
1982 1994 2001 2007 2011 2013 2015 2016 2018 2019 2020 2021 2022 2023 2024 2025

How Many Qubits to Break RSA-2048?

O(1000) error-free quantum bits to break RSA-2048



Equivalent to 2^{1001} classical bits



2^{265} atoms in the visible universe

How Many Qubits to Break RSA-2048?

The Pinnacle Architecture: Reducing the cost of breaking RSA-2048 to 100 000 physical qubits using quantum LDPC codes

Paul Webster,* Lucas Berent, Omprakash Chandra, Evan T. Hockings,
Nouédyn Baspin, Felix Thomsen, Samuel C. Smith, and Lawrence Z. Cohen†
Iceberg Quantum, Sydney

The realisation of utility-scale quantum computing inextricably depends on the design of practical, low-overhead fault-tolerant architectures. We introduce the *Pinnacle Architecture*, which uses quantum low-density parity check (QLDPC) codes to allow for universal, fault-tolerant quantum computation with a spacetime overhead significantly smaller than that of any competing architecture. With this architecture, we show that 2048-bit RSA integers can be factored with less than one hundred thousand physical qubits, given a physical error rate of 10^{-3}, code cycle time of 1 μs and a reaction time of 10 μs. We thereby demonstrate the feasibility of utility-scale quantum computing with an order of magnitude fewer physical qubits than has previously been believed necessary.

Shor's algorithm is possible with as few as 10,000 reconfigurable atomic qubits

Madelyn Cain[1,*,†], Qian Xu[1,2,*,‡], Robbie King[1], Lewis R.B. Picard[1], Harry Levine[1,3],
Manuel Endres[1,2], John Preskill[1,2], Hsin-Yuan Huang[1,2], Dolev Bluvstein[1,2,§]
[1] *Oratomic, Pasadena, California 91125, USA*
[2] *California Institute of Technology, Pasadena, California 91125, USA*
[3] *Department of Physics, University of California, Berkeley, California 94720, USA*
[†] *mcain@oratomic.com,* [‡] *qxu@oratomic.com,* [§] *dbluvstein@oratomic.com*
[*] *These authors contributed equally*

(Dated: March 31, 2026)

Quantum computers have the potential to perform computational tasks beyond the reach of classical machines. A prominent example is Shor's algorithm for integer factorization and discrete logarithms, which is of both fundamental importance and practical relevance to cryptography. However, due to the high overhead of quantum error correction, optimized resource estimates for cryptographically relevant instances of Shor's algorithm require millions of physical qubits. Here, by leveraging advances in high-rate quantum error-correcting codes, efficient logical instruction sets, and circuit design, we show that Shor's algorithm can be executed at cryptographically relevant scales with as few as 10,000 reconfigurable atomic qubits. Increasing the number of physical qubits improves time efficiency by enabling greater parallelism; under plausible assumptions, the runtime for discrete logarithms on the P-256 elliptic curve could be just a few days for a system with 26,000 physical qubits, while the runtime for factoring RSA-2048 integers is one to two orders of magnitude longer. Recent neutral-atom experiments have demonstrated universal fault-tolerant operations below the error-correction threshold, computation on arrays of hundreds of qubits, and trapping arrays with more than 6,000 highly coherent qubits. Although substantial engineering challenges remain, our theoretical analysis indicates that an appropriately designed neutral-atom architecture could support quantum computation at cryptographically relevant scales. More broadly, these results highlight the capability of neutral atoms for fault-tolerant quantum computing with wide-ranging scientific and technological applications.

O(1000) error-free quantum bits to break RSA-2048

Equivalent to 2^{1001} classical bits

2^{265} atoms in the visible universe

Theory: 100,000 "Noisy" Qubits to Break RSA-2048

Experiment: Largest commercial QCs have ~1,000 qubits with MUCH larger noise (error rates)

Opportunities

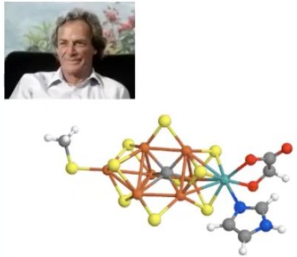
[Credit: Scott Aaronson's Public APS Talk]

Caveat: "QC works" ≠ "QC speeds up your application"!!

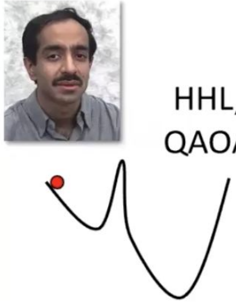
Finding clear real-world applications of QC, beyond the "Big Three," remains as much of a challenge as ever

1. Breaking Current Public-Key Cryptography

2. Simulating Quantum Physics and Chemistry



3. Everything Else...



HHL, QAOA

Many Topics To Discuss…

Week 1

- Qubits, gates, and circuits
- Entanglement and Quantum Teleportation
- First example of quantum advantage: Simon’s algorithm

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- Quantum Fourier Transform
- Quantum Phase Estimation
- Shor's Factoring Algorithm

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Week 3 (Up to you…)

- Discrete Log Algorithm
- Grover’s Search Algorithm
- Quantum Encryption via BB84
- Quantum Optimization Algorithms

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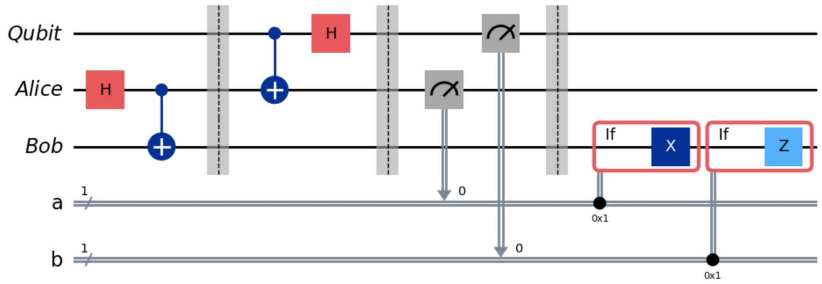
Qiskit



Ground Rules

1) **Let's have fun!**

2) **Help each other**
    a) Ask questions
    b) Be engaged

3) **Do the homeworks (due before each following lecture)**
    a) Collaboration will be necessary
    b) Explore, explore, explore

4) **The use of AI is allowed and encouraged**
    a) Document your use (prompts, responses, etc.)
    b) Write up your own solution – handwritten when it is not code

INTRODUCTION TO QUANTUM BITS AND GATES

Classical and Quantum Bits

Classical Bits

Binary value of 0 or 1

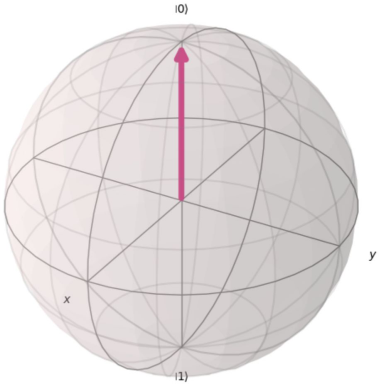
Classical and Quantum Bits

Classical Bits

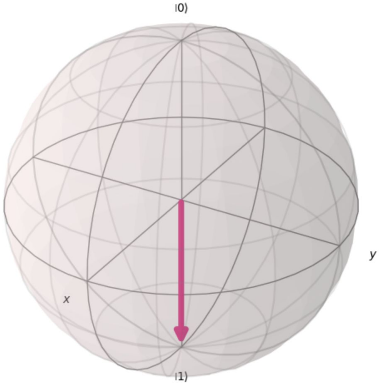
Binary value of 0 or 1

Quantum bits (qubits)

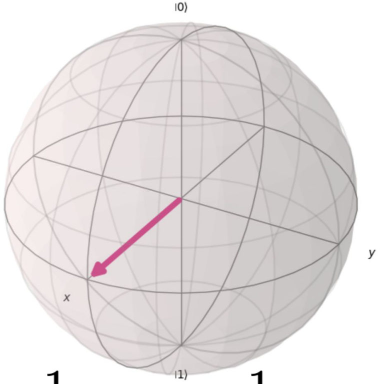
"Superposition state" $|\psi\rangle = a|0\rangle + b|1\rangle$



$|0\rangle$



$|1\rangle$



$$\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

Reading from a Qubit (Measurement)

The quantum state of a single qubit is not directly readable

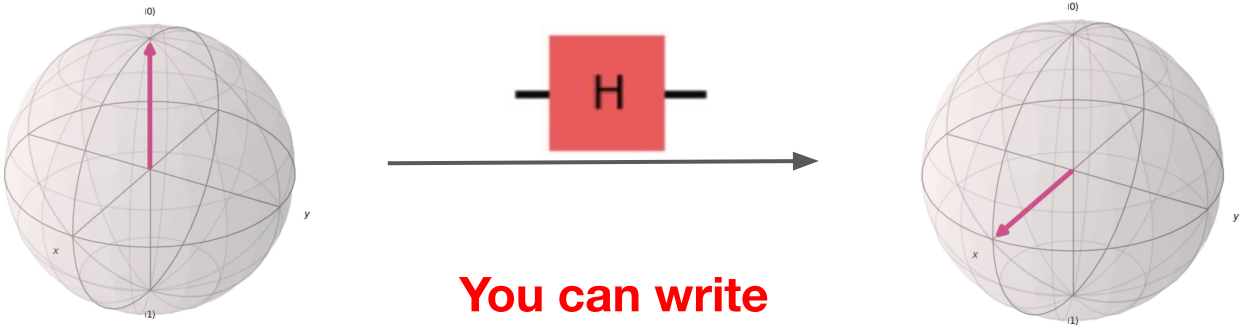
- When you measure a qubit, you can only read a classical bit value of 0 or 1
  - The quantum state “collapses” to 0 or 1

- Suppose you have many qubits with the same quantum state $|\psi\rangle = a|0\rangle + b|1\rangle$

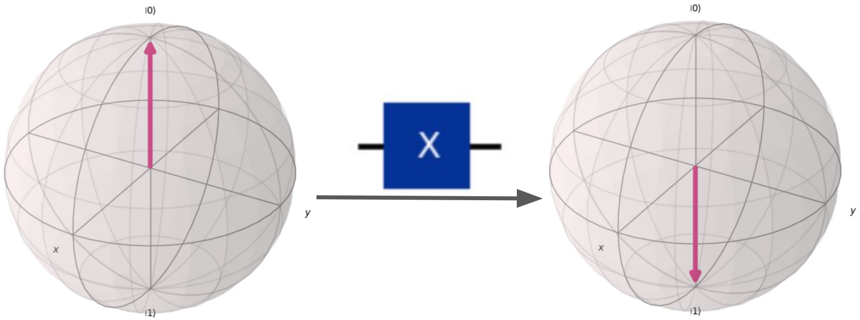
- Measure all of them – you will get roughly
  - $|a|^2$ number of 0’s
  - $|b|^2$ number of 1’s

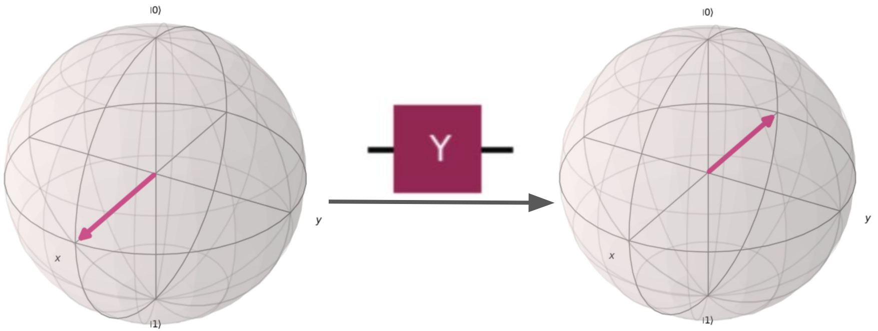
- Quantum mechanics is probabilistic ⇒ Learning the state of a qubit with good accuracy requires a LOT of identical qubits!

Writing to a Qubit (Quantum Gates)

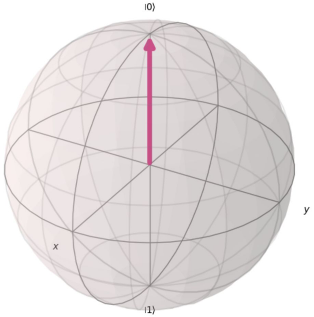


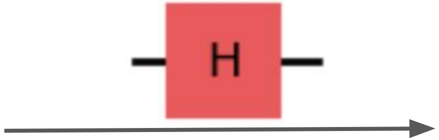
You can write information to qubits using quantum gates

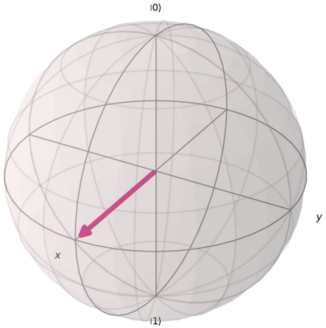




Writing to a Qubit (Quantum Gates)







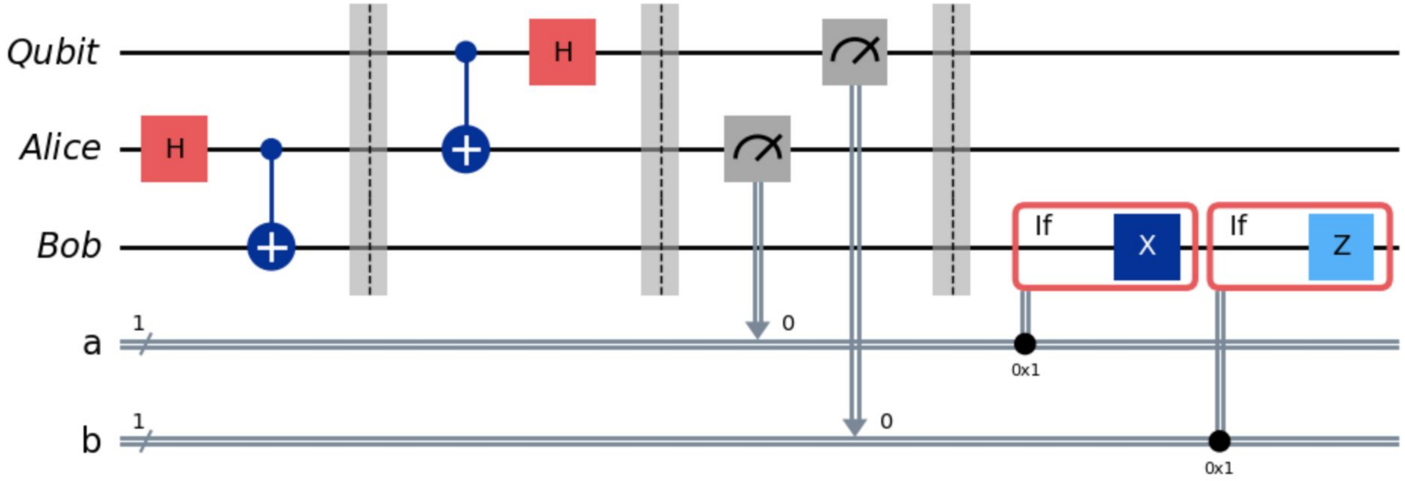
$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$H|0\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

Quantum gates are just matrix operations on quantum state-vectors

Quantum Circuits



You can manipulate quantum information using circuits, a sequence of quantum gates acting on multiple qubits

Quantum gate operations on multiple qubits
⇒ Transfers and entangles information between qubits

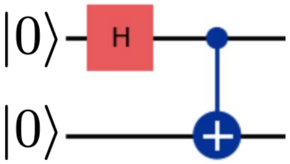
Entanglement

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

- **If we measure one qubit to be 0 ⇒ The other qubit must also be 0**
- **If we measure one qubit to be 1 ⇒ The other qubit must also be 1**

Two entangled qubits contain the same information

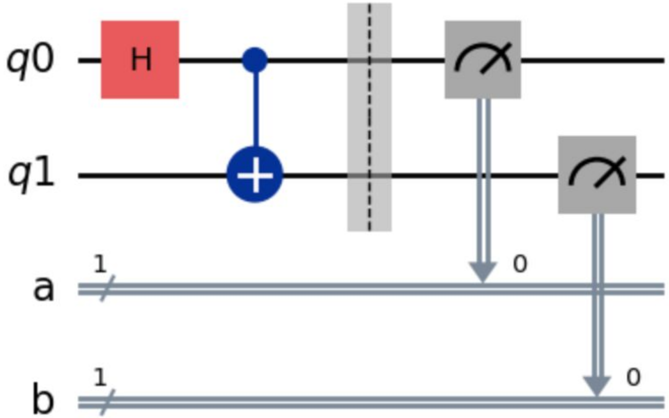
```python
from qiskit import QuantumCircuit, QuantumRegister

qubit0 = QuantumRegister(1, "qubit_0")
qubit1 = QuantumRegister(1, "qubit_1")

protocol = QuantumCircuit(qubit0, qubit1)

# Perform gate operations that entangle qubits
protocol.h(qubit0)
protocol.cx(qubit0, qubit1)

display(protocol.draw(output="mpl"))
```

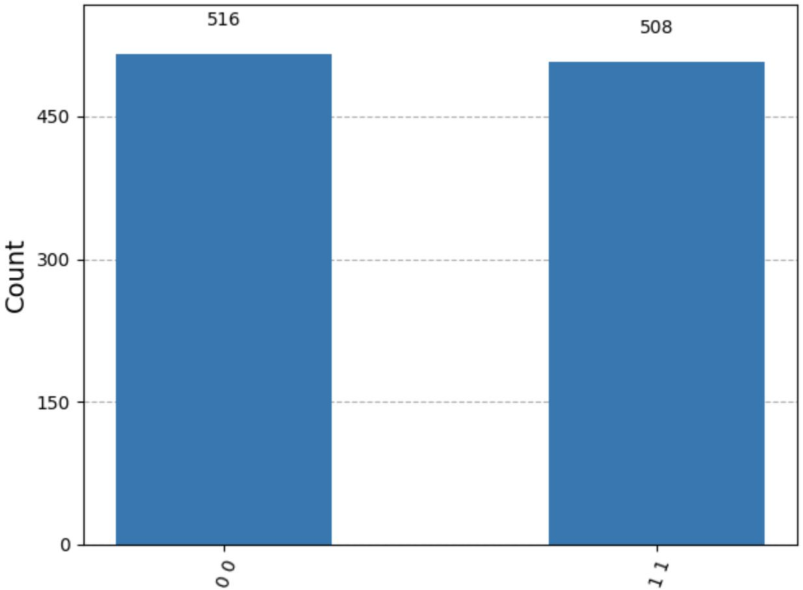


$$|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Entanglement in Practice

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

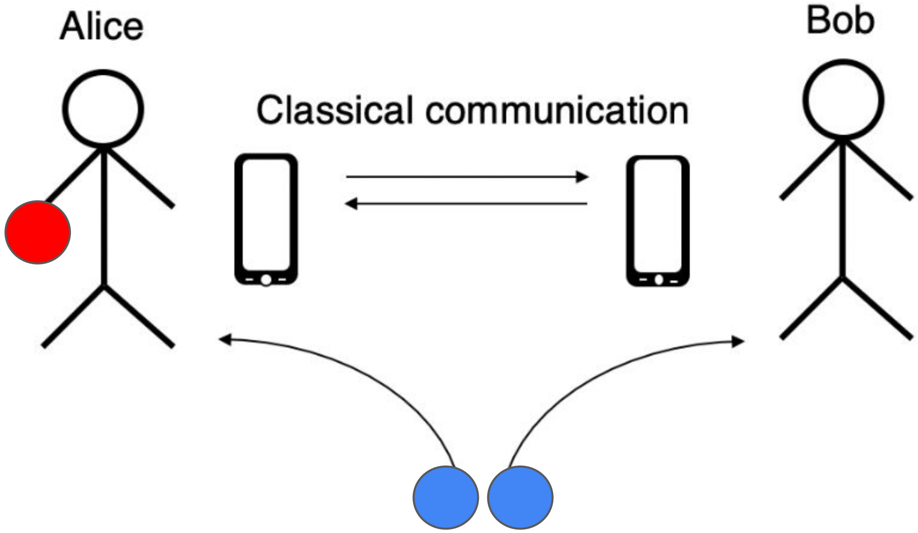




IBM's Simulator

Quantum Algorithms and Teleportation

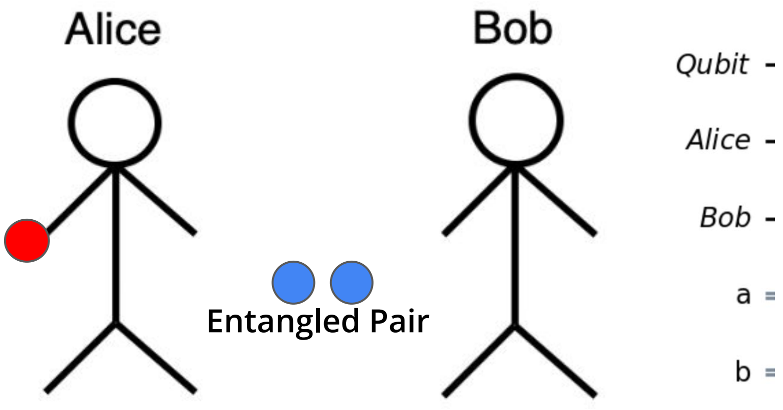
Teleportation – Quantum Information Transfer



Entangled qubits

- Alice transfers a qubit state to Bob using entanglement
- Neither Alice nor Bob knows the state of this qubit
- This protocol requires Alice to communicate classical information to Bob (0s and 1s)

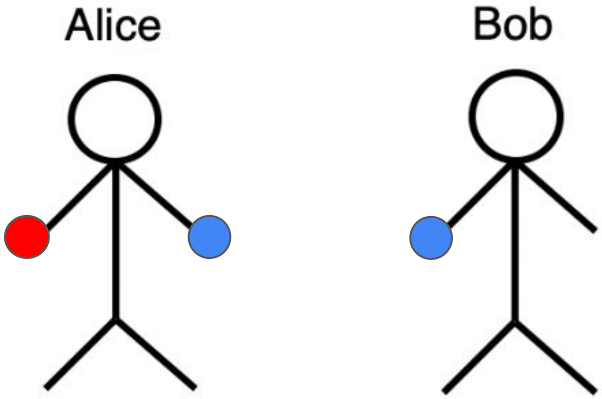
Initial State Preparation

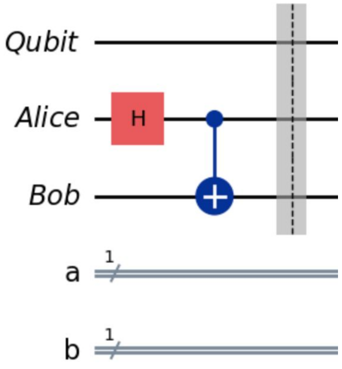



```python
qubit = QuantumRegister(1, "Qubit")
ebit0 = QuantumRegister(1, "Alice")
ebit1 = QuantumRegister(1, "Bob")
a = ClassicalRegister(1, "a")
b = ClassicalRegister(1, "b")
protocol = QuantumCircuit(qubit, ebit0, ebit1, a, b)
```

[Reference: IBM Quantum Learning]

Shared Entanglement

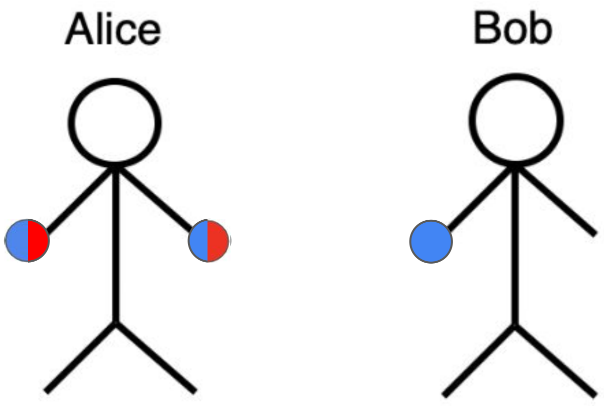


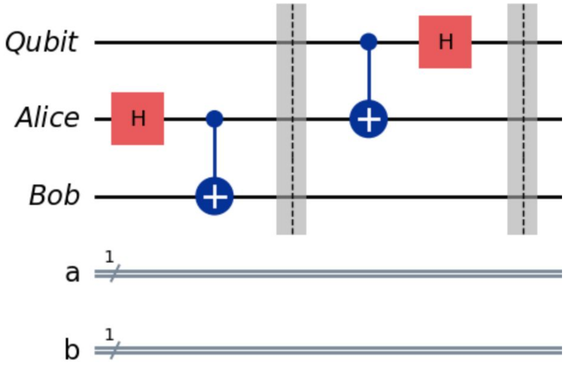



```python
# Prepare entangled qubits used for teleportation
protocol.h(ebit0)
protocol.cx(ebit0, ebit1)
protocol.barrier()
```

[Reference: IBM Quantum Learning]

Alice Entangles Her Qubit

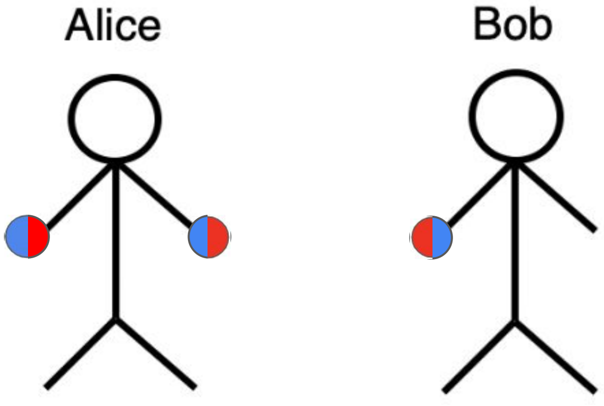


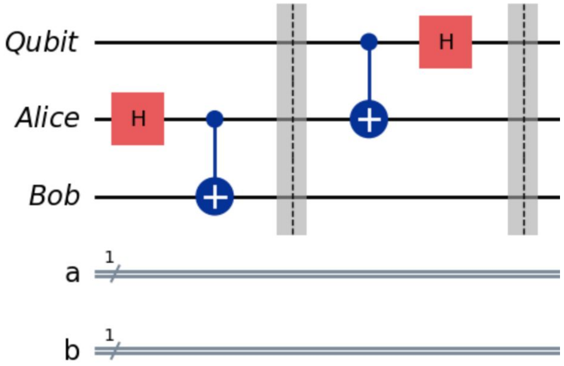



```
# Alice's operations
protocol.cx(qubit, ebit0)
protocol.h(qubit)
protocol.barrier()
```

Reference: IBM Quantum Learning

Entanglement ⇒ Information Transferred to Bob

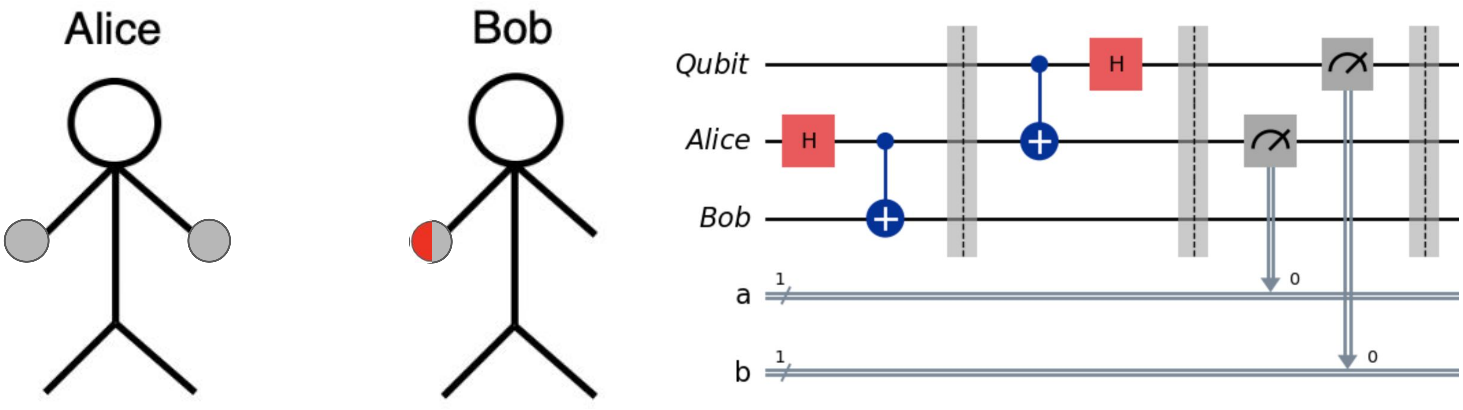





```
# Alice's operations
protocol.cx(qubit, ebit0)
protocol.h(qubit)
protocol.barrier()
```

Reference: IBM Quantum Learning

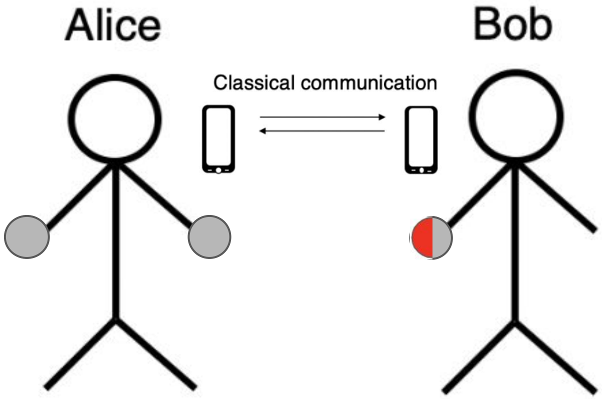
Alice Measures Her Qubits

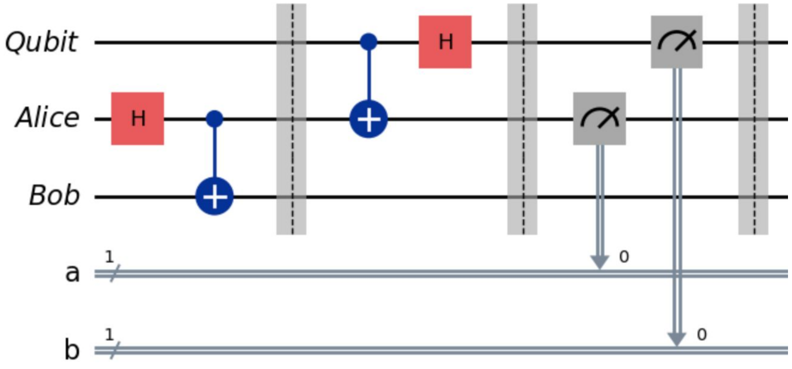



```python
# Alice measures and sends classical bits to Bob
protocol.measure(ebit0, a)
protocol.measure(qubit, b)
protocol.barrier()
```

[Reference: IBM Quantum Learning]()

Alice Communicates Her CLASSICAL Info to Bob

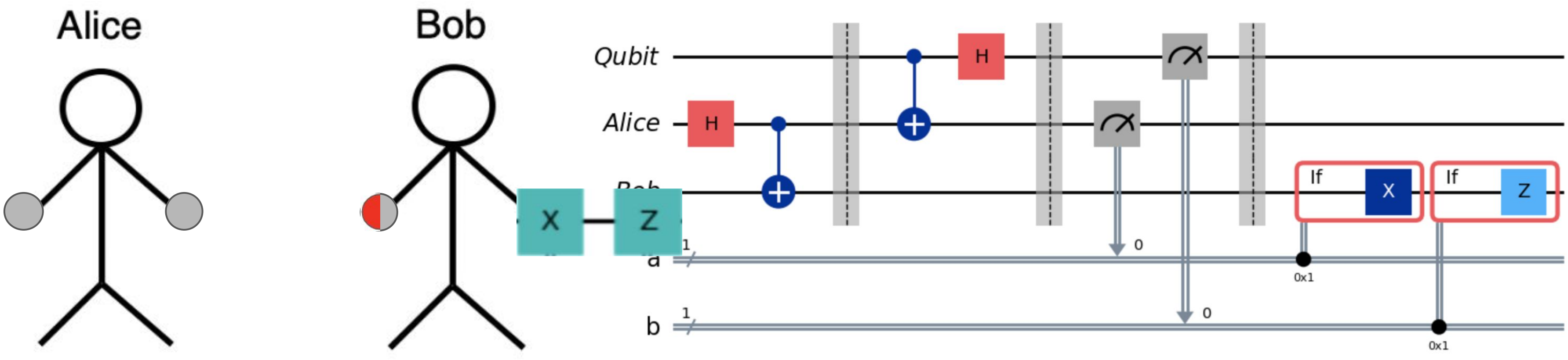





```
# Alice measures and sends classical bits to Bob
protocol.measure(ebit0, a)
protocol.measure(qubit, b)
protocol.barrier()
```

[Reference: IBM Quantum Learning]()

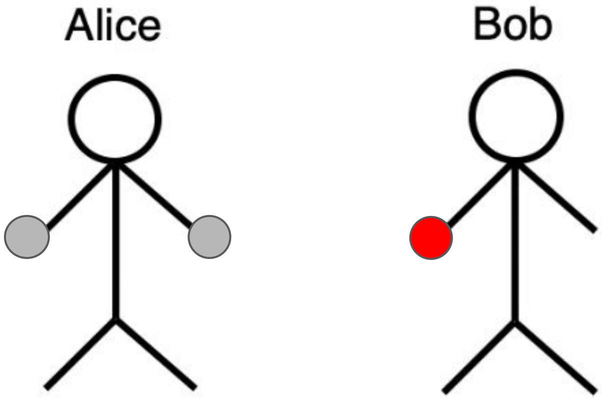
Bob Alters His Qubits Based on Alice's Measurements

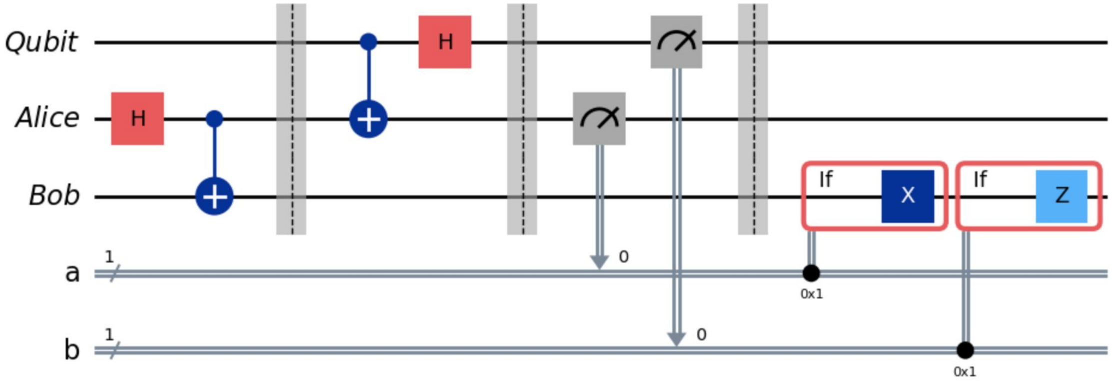



```
# Bob uses classical info to conditionally apply gates
with protocol.if_test((a, 1)):
   protocol.x(ebit1)
with protocol.if_test((b, 1)):
   protocol.z(ebit1)
```

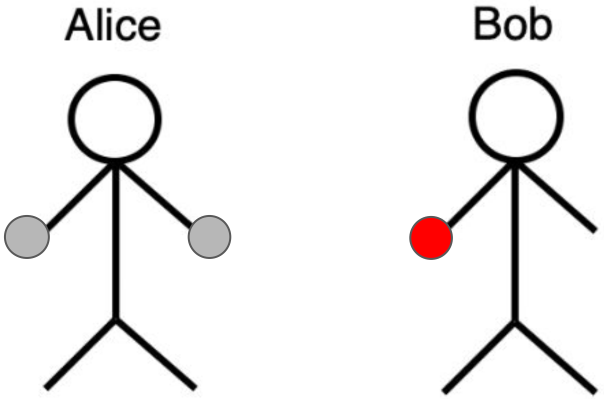
[Reference: IBM Quantum Learning]

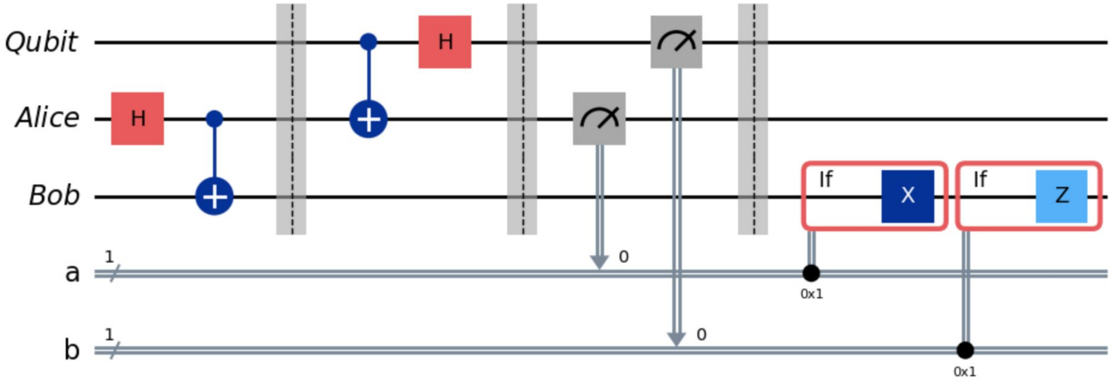
Bob Recovers Alice’s Original Qubit State





Takeaways From Teleportation



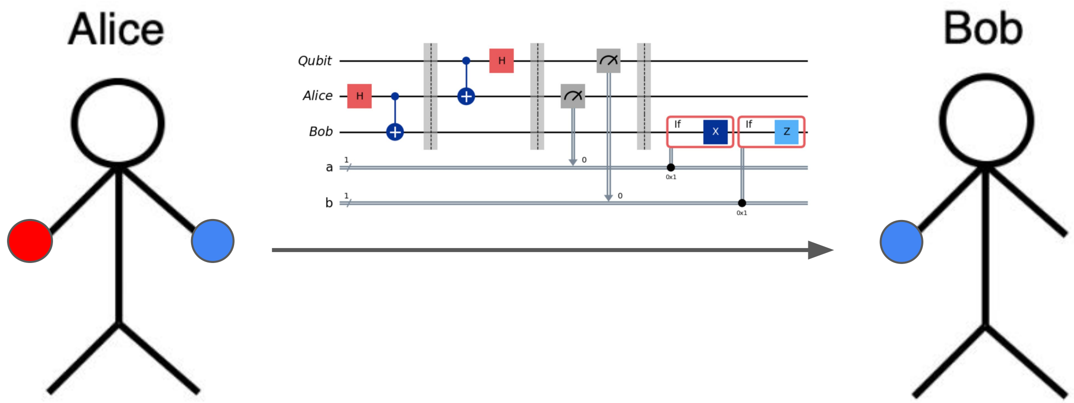


- **Neither Alice nor Bob needed to know the initial qubit state**
  ⇒ Requires no knowledge of the information being transferred

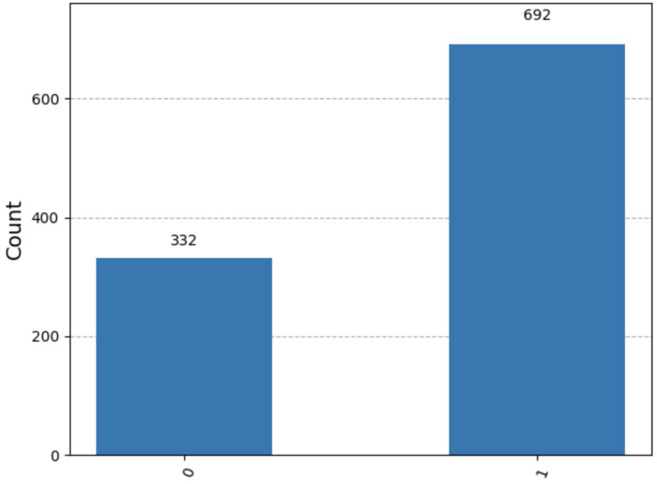
- **Only classical information is transferred between Alice and Bob**
  ⇒ An eavesdropper would gain no useful quantum information

- **Alice's original qubit state was transferred, NOT cloned, onto Bob's qubit**

Teleportation in Practice



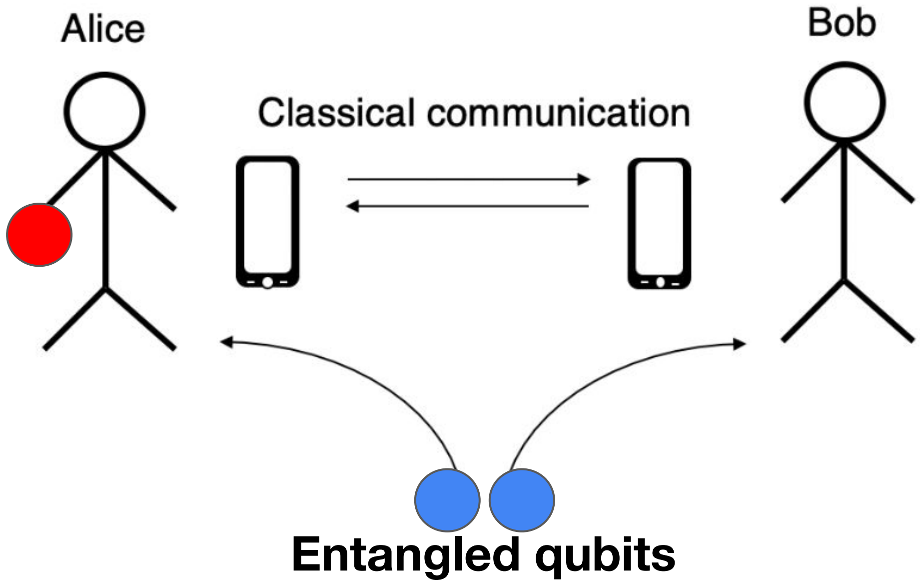
$$|\psi\rangle_{\text{Alice}} = \frac{1}{\sqrt{3}}|0\rangle + \sqrt{\frac{2}{3}}|1\rangle$$



IBM's Simulator

BACKUP

Teleportation – Quantum Information Transfer



Initial States
- Alice has one qubit
- Alice and Bob share a pair of entangled qubits

Information Entanglement
- Alice entangles her qubit with one qubit in the pre-entangled pair
- The entanglement transfers onto the other qubit in the pre-entangled pair, held by Bob

The Final State
- Alice communicates measurements of her qubits to Bob
- Bob uses that information to recover Alice's original qubit state